

Development of a Data-driven Self-adaptive Upper Limb Virtual Rehabilitation System for Post Stroke Elderly

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ABSTRACT

The study aims to develop a virtual rehabilitation system to assist upper limb motor training for older post-stroke patients. The system contains data-driven virtual exergames simulating the task-oriented training; receives the rehabilitation prescription and the online data collected from the multi-mode hand controller and the depth camera; assesses online patient's performance which in turn updates the data of virtual exergames to adapt to the patient's training progress. Its innovation is providing precision rehabilitation via gaining the personalized learning experience to improve the adherence to and effectiveness of virtual rehabilitation.

Keywords: Virtual rehabilitation, precision rehabilitation, virtual exergame, machine learning.

Index Terms: I.3.7 [Three-Dimensional Graphics and Realism]: Virtual Reality

1 INTRODUCTION

Stroke is the major cause of adult-acquired disability worldwide, and more than 80% of them reportedly present with upper limb (UL) motor impairment [1]. Recent studies have proven that virtual reality-based rehabilitation, virtual rehabilitation (VR), can positively facilitate motor learning in UL rehabilitation [2]. Significantly, the task-orientated gamification and the multi-modal feedback of VR can motivate stroke patients to engage in therapy and lead to positive reinforcement in promoting functional improvements in UL.

VR has gained more importance and application in offering telerehabilitation at home or in the community. However, since most VR design emphasizes the “just-right-challenge” [3] gamification to engage patients at the beginning of rehabilitation, VR usually could not collect patients' performance data by adapting to patients' training achievements. Hence, patients might gradually lose interest in VR. A recent study proposed a data-driven method to evaluate changes in motor function in long-term rehabilitation [4], which shed light on designing a VR to support precision rehabilitation for post-stroke elderly.

Moreover, the adaption of motor training in VR should consider the hyper-adaptability [5] of neural networks and motor control. A study [6] finds that the adaption inertia and proprioceptive drift could happen when the movement of a virtual limb is not matched with the actual limb's movements. The finding reveals the challenge in designing the appropriate motor training in VR to adapt to the impaired dynamics of body parts, especially when

embodying limb motions on a virtual avatar.

2 VIRTUAL REHABILITATION SYSTEM

2.1 System Architecture

The system architecture is shown in Figure 1, which has three data layers and three core modules to process the rehabilitation data.

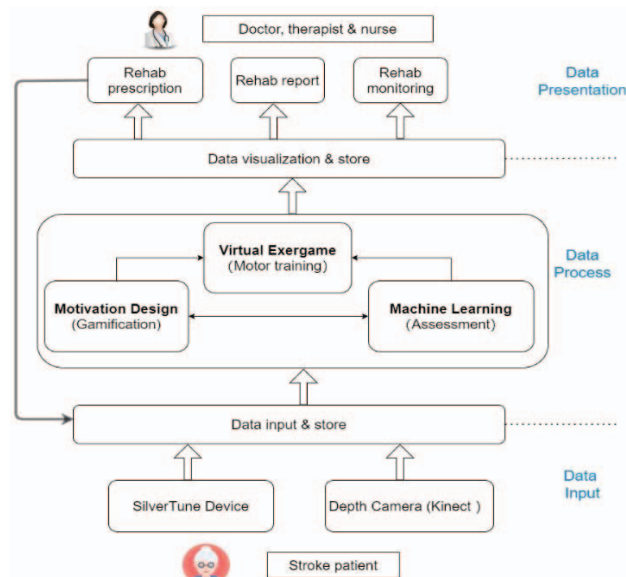


Figure 1: The architecture of the virtual rehabilitation system.

2.1.1 Data Input Layer

The data input layer comprises the offline rehabilitation prescription and online data inputted from the self-design multi-mode hand controller and the depth camera, Microsoft Kinect camera. The offline rehabilitation prescription could be used to select the type of virtual exergame and set up the exergame parameters, such as the duration and difficulty level of the virtual exergame.

The hand controller shown in Figure 2 is equipped with an accelerometer, sixteen tact switch sensors, a MCU, etc., in which MCU sends sensor data to a personal computer via Bluetooth. The shape of the controller can be changed into various modes, such as the triangle shape and the L shape, to assist patients in playing multiple visual exergames to achieve the training target. The Kinect camera could track the patient's skeleton to measure the motion data of the upper limb and the trunk in the present system. The motion data are used to assess the patient's UL performance, which is processed in the machine learning module in the data process layer.

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2.1.2 Data Process Layer

The virtual exergame module simulates task-oriented training to assist patients' rehabilitation and satisfy the requirements of rehabilitation prescription. The design of virtual exergames, shown in Figure 2, adopts the data-driven approach to online customize its content based on the data collected in the data input layer and the input from the other two modules in the same data process layer. The online customization ensures that the training in a virtual scene could adapt to the patient's rehabilitation progress. Moreover, the virtual exergame is played by receiving input from the multi-mode hand controller rather than from the Kinect camera. Thus, the virtual exergame does not embody the patient's UL motion on an avatar to avoid some potential risks, such as adaption inertia [6], observed in previous VR.

The machine learning module processes the data collected from the data input layer. Currently, a support vector machine (SVM) is designed to classify the Fugl Mayer Assessment (FMA) level based on the collected motion data of the upper limb, which could assess the patient's training performance. The assessment results could be inputted into the other two modules of the same layer. For example, the assessment results could be used to refine the training parameters of a virtual exergame, such as the difficulty level of a training task.

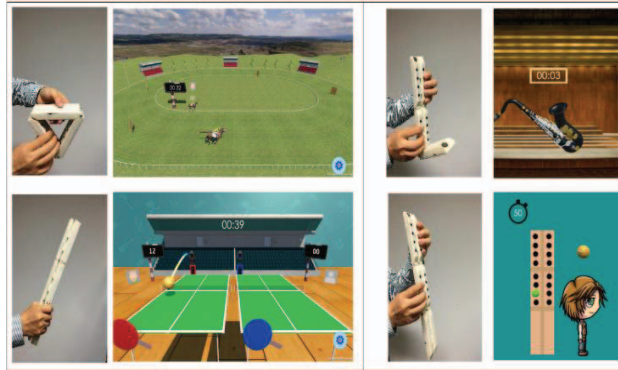


Figure 2: The self-design multi-mode hand controller, two motor exergames (left: horse racing and ping pang games) for the upper limb, and two motor exergames (right: saxophone playing and ball collection) for hand fingers.

The motivation design module deploys the gamification theory to design the virtual exergame to promote patient adherence to the training. Precision rehabilitation cannot achieve the expected training target without the patient's good adherence. The inherent gaming experience, such as horse-running competition, could greatly promote patient motivation. The gamification is also moderated by the assessment results inputted from the machine learning module.

2.1.3 Data Presentation Layer

The rehabilitation prescription is the plan designed by clinicians to guide patients to correctly take rehabilitation sessions, which could be inputted into the data input layer to initialize the exergame. The rehabilitation report summarizes the patient's progress in past rehabilitation sessions. The rehabilitation monitoring module visualizes patients' ongoing performance and historical progress and alerts clinicians and patients about the assessment results.

2.2 Data-driven Virtual Exergame

There are four data models: character, coordinator, gameplay, and context. The character defines the property and behavior of all

game elements, and one unique character is the avatar describing the stroke patient's profile. The coordinator defines training requirements and assessment results. The gameplay defines the set of actions the character could perform to play the exergame to achieve the training target. The context defines the gaming context, such as the virtual scene for upper limb training.

The above four data models are implemented in the data-driven virtual exergame consumes. The data flow starts from the coordinator receiving the rehabilitation prescription and online assessment results, and then the coordinator updates the contents of both gameplay for the required training movement and context to provide the appropriate scene for training. The setup of training movement and scene will then define the character's behavior in the virtual exergame, which guides the patient's movements captured by the hand controller to trigger the character to move in the virtual scene. The patient's movements would then be assessed online to be inputted to the coordinator, which adapts to the patient's training progress.

3 PILOT TEST

A pilot test was conducted to evaluate the efficacy of the developed system with eleven stroke patients (mean age 69.5 years) in one rehabilitation center. The initial results showed improved satisfaction ($p=0.007$) and perceived performance ($p=0.005$) between the pre- and post-intervention. However, there was no significant difference in the FMA test ($p>0.05$).

4 CONCLUSION

A data-driven self-adaptive virtual rehabilitation is developed to assist the upper limb motor training. More clinical trials are needed in the next stage of development to improve the precision of providing adaptive virtual rehabilitation.

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